Advisor brief: the revised combined empirical paper and a proposed method scope

16 October 2026. Revision of the combined manuscript at commit `3b5065805`; the later `PCRL_Combined_Empirical_Revised` files were not available on this machine and were not revised.

1. What the paper now asks

For stored representation models and published defenses, what can a recipient recover about a declared protected attribute from everything it receives, alone or linked with one other recipient, and how useful is that release for the recipient's task? The paper now argues from the problem to the evidence. Protection methods make promises scoped to an object and an attacker class: LEACE covers linear classifiers reading the representation, and FARE certifies classifiers that read its released cell. A recipient may receive more than that object, namely a task output and, if recipients link records, another recipient's output. The results separate three outcomes: a method failing its own test, a test holding while recovery occurs outside its scope, and recovery through a surface the test does not cover.

2. Why this adversary and this access

The claims concern a release policy, so we chose to evaluate an adversary that holds what the policy delivers. An authorized recipient necessarily possesses its own release, so that part describes the service rather than an assumption. Two capabilities are granted, not established. *Linkage*: two recipients serving the same person may pool releases by record; we grant perfect linkage, the attacker-favourable limit, without assuming that governance prevents linkage or that pooling is unavoidable. *Labelled auxiliary data*: our attackers are supervised and learn from releases of other records with the attribute, which models a recipient that knows the attribute for some of the people it serves; availability, size and population match are assumptions. A stronger attacker alone does not justify more access. Inputs held by the recipient, published parameters and permitted repeated queries are distinct threat models that need not involve a breach; insiders are another distinct threat model. All are untested, not dismissed. Within the evaluated access a stronger attacker could recover more, so each recovery figure is a lower estimate, not a guarantee. The paper now has a table that states, for each experiment, what the recipient receives, the labelled sample (5,388 to 7,571 labelled rows for fitting and 2,082 to 2,239 for selection, depending on dataset and experiment), how attackers are fitted, selected and scored, what they know about the defense, how releases are linked and what is untested. It also reports two facts the earlier draft omitted: the noise scale and FARE's configuration were selected against our own attacker slate on its validation rows, and two exploratory attackers that know the noise scale recovered more from the noisy features than the slate did: the first, within the evaluated access, 0.546 on Adult and 0.526 on HMDA; the second, fitted on clean auxiliary representations that no contract releases, 0.563 and 0.575; the slate 0.528 and 0.513 (exploratory, no paired test).

3. What the results establish

- **Scope, not failure.** Official LEACE meets its linear bound on every map, while nonlinear attackers recover the attribute about as well as from untreated features (AUC 0.817 and 0.863 against 0.815 and 0.870).
- **A method's own test failing.** The encoders' stored linear check fails on 11 of 42 combinations of purpose–attribute pair and encoder seed (the earlier manuscript said 12; the committed table and the source's own list give 11).
- **Other surfaces.** Releasing the untreated model's output beside noisy or FARE features restores recovery to 0.76–0.79, and outputs reveal more than the true label alone would.
- **Utility changes the reading.** With useful heads, FARE's score recovers 0.539 against 0.773 untreated, but a same-budget tree without the fairness term already reaches 0.648, and FARE's accuracy is not shown to stay within one point. On an easier task, compression reproduces most of FARE's effect.
- **Linked recipients.** In the one coalition evaluated (Adult income and occupation-group recipients, race), two recipients granted perfect linkage recover race better than either alone in every output format, including hard decisions (0.611 against 0.576).

All of this is development evidence on previously used rows, with intervals conditional on the fitted heads and attackers.

4. What remains unproved

There is no population privacy guarantee: recovery is a property of our attacker slate and data budget. Differences in recovery may reflect information removed, limits of our attackers, or both, and the paper does not separate them. A deterministic release need not be reversible, and the paper does not establish whether any release keeps or discards the attribute's posterior. The excluded threat models, imperfect linkage and other auxiliary budgets are untested. Nothing is fresh confirmation, and there is no new method and no composition or synergy result.

5. Proposed objective for the future method

Develop purpose-specific releases that retain the usefulness required by each permitted task while reducing sensitive-attribute recovery from everything recipients receive, both individually and when releases are linked. The endpoint is empirical: recovery under declared access, attacker families and auxiliary-data budgets, reported beside the utility of what is released. A mechanism may aim to remove information that the tasks do not need and still be judged by finite attacks; the two goals are compatible. A claim about arbitrary attackers would need a theorem; exact examples demonstrate their own laws, not real-population privacy. A competitive method need not be a joint design, and joint benefit, competitiveness and novelty are separate questions. Existing utility caps, release contracts and references stay unchanged. No competitive new mechanism has been established, and agreeing on this scope does not supply one.

6. Feedback and decisions needed

1. Is the question and the stated threat model the right framing for the empirical paper, and is the per-experiment access table sufficient?
2. Should the paper stay at its current evidence cutoff? Possible extensions are listed separately and none is proposed for immediate work.
3. What usefulness does the service actually need: calibrated probabilities, ranking scores, or decisions? This choice defines the method problem.
4. Which attacker families, auxiliary budgets, linkage model and comparators should a future method be judged against, and is a formal guarantee a goal?